

DIDO: Another alternative

Distributed-input distributed-output (DIDO) is a new radio technology from Steve Perlman and the Rearden Companies.

"DIDO is an attempt to provide another alternate architecture in cellular communications similar to what cloud computing is doing in the processing and database domains," explains Dr Borkar. It is an attempt to offload the user device and provide direct communications from a data-centre based array of transmitting devices.

"Its hype is currently misleading in that it seems to squeeze more bandwidth magically when you add users in a cellular system. Like all other technologies, it still needs to obey the physical laws of communications channel limitations. It attempts to overcome such limitations by providing multiplexed dedicated channels to individual users to give appearance of providing total bandwidth of the cell to each of the users in the coverage area. Clearly, there is impact on latency and other Quality of Service (QoS) parameters, which may not meet real-time application needs. Since only limited details are currently available in the literature, it is premature to compare DIDO with any of the existing technologies," comments Dr Borkar.

and devices identified and made available along with the infrastructure and related entities before it can be used widely. Some limited prototype-friendly deployments have taken place in the last year or so but the availability of receiving devices that require arrays of photodiodes is still limited.

"The technology also needs to evolve in many areas of miniaturisation including application-specific integrated circuits (ASICs), optical devices including photodiodes, compact arrays and modulation of optical signals. Another major area of concern is the hardware needed whenever we interface optics with electronics," says Dr Borkar.

This concurs well with Dr Povey, who says: "The receiver optics and sensor is always an issue. We have to consider how to collect enough optical power using a non-ideal positioning of the sensor. The technology can be improved a lot. Also, we are trying to miniaturise the technology, which is challenging."

A few other VLC projects

Li-Fi is just one of today's hopeful VLC technologies. Research in VLC has been going on in the UK, the USA, Germany, Korea and Japan since 2003. Casio, Intel, Samsung and Boeing are some of the biggies that are active in this field.

Some years ago, Tokyo-based Nakagawa Laboratories demonstrated underwater visible light communication technology for scuba divers. Later, in

December 2010, an American company called LVX deployed light-powered broadband services at six buildings in St Cloud, Minnesota. However, they were able to achieve a speed of only 3 Mbps, which is quite slow as per today's broadband standards.

Around the same time, a team of researchers from Siemens and Fraunhofer Institute for Telecommunications demonstrated transmission of 500 Mbps over a distance of 5 metres using a single white LED, and transmission of 100 Mbps over a longer distance using five LEDs.

Other ongoing efforts in this space include Boston University's Smart Lighting Engineering Centre, the European Union's OMEGA-Home Gigabit Access project and University of California's Ubiquitous Communication by Light (UC-Light) Centre at Bourns College of Engineering.

Quite recently, at the *Consumer Electronics Show 2012*, Casio demonstrated the prototype of a VLC product. Casio has been a member of the Visible Light Communications Consortium since it was initiated in 2004. In addition to fundamental research and standards development for VLC, Casio is developing applied technology for receiving signals through the use of image sensors such as complementary metal-oxide-semiconductor (CMOS) and charge-coupled-device (CCD) sensors.

Casio's image sensor communications technology can determine the point of data transmission while simultaneous-

ly receiving many signals. The company has used its image sensor communications technology to develop the prototype of a smartphone VLC system for consumer and commercial applications. The system flashes smartphone screens to achieve VLC. Casio demonstrated a potential social media application.

This is how the system works: When someone takes a photo with a smartphone camera, the subjects simply turn the screens of their smartphones toward the camera device to display personal information or messages in the photo. The photo-taker's phone can receive data from up to five smartphones to add information to the photo. The information is displayed in message balloons of up to 120 characters, with customisable balloon shapes and image frames. Information such as e-mail addresses, telephone numbers and social network usernames is automatically saved on the photo-taker's smartphone. Twitter-upload tweets the image containing the messages. The same technology can be extended to backlit ad banners and more.

Overall, the developments are quite interesting, and we could hope to see Li-Fi and other VLC technologies in the mainstream five years down the line.

However, one misconception needs to be cleared at the very outset: Li-Fi or similar technologies, for that matter, do not compete with Wi-Fi, but complement it.

"I don't see a battle between Li-Fi and Wi-Fi," says Dr Povey. "Cellular data does not really compete with Wi-Fi. They complement each other. I see the same position with Li-Fi. When the cellular networks became congested, we were encouraged to use Wi-Fi at home or in the office to offload the excess demand. Now Wi-Fi is getting overloaded and so for short-range high-data rate links, it seems logical to offload the excess demand to Li-Fi. Because of the exponentially growing demand for data, we have a choice—a light bulb or a cable," he adds. ●

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